

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. - 11



By - Abhishek Sir



Recap of Previous Lecture



Topic

CIDR

Topic

IP Configuration



Topics to be Covered



Topic

IP Configuration

Topic

ARP

(Imp.)

ABOUT ME



Hello, I'm **Abhishek**

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[MCQ]

[GATE-2024][2 Mark]



#Q. Which one of the following [CIDR prefixes] exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

- ☒ A 10.12.2.0/23
- ☐ B 10.12.2.0/24
- ☒ C 10.12.0.0/22
- ☐ D 10.12.2.0/22

Ans: A

10.12.2.0 : 10.12.00000001 0.00000000

10.12.3.255 : 10.12.00000001 1.11111111

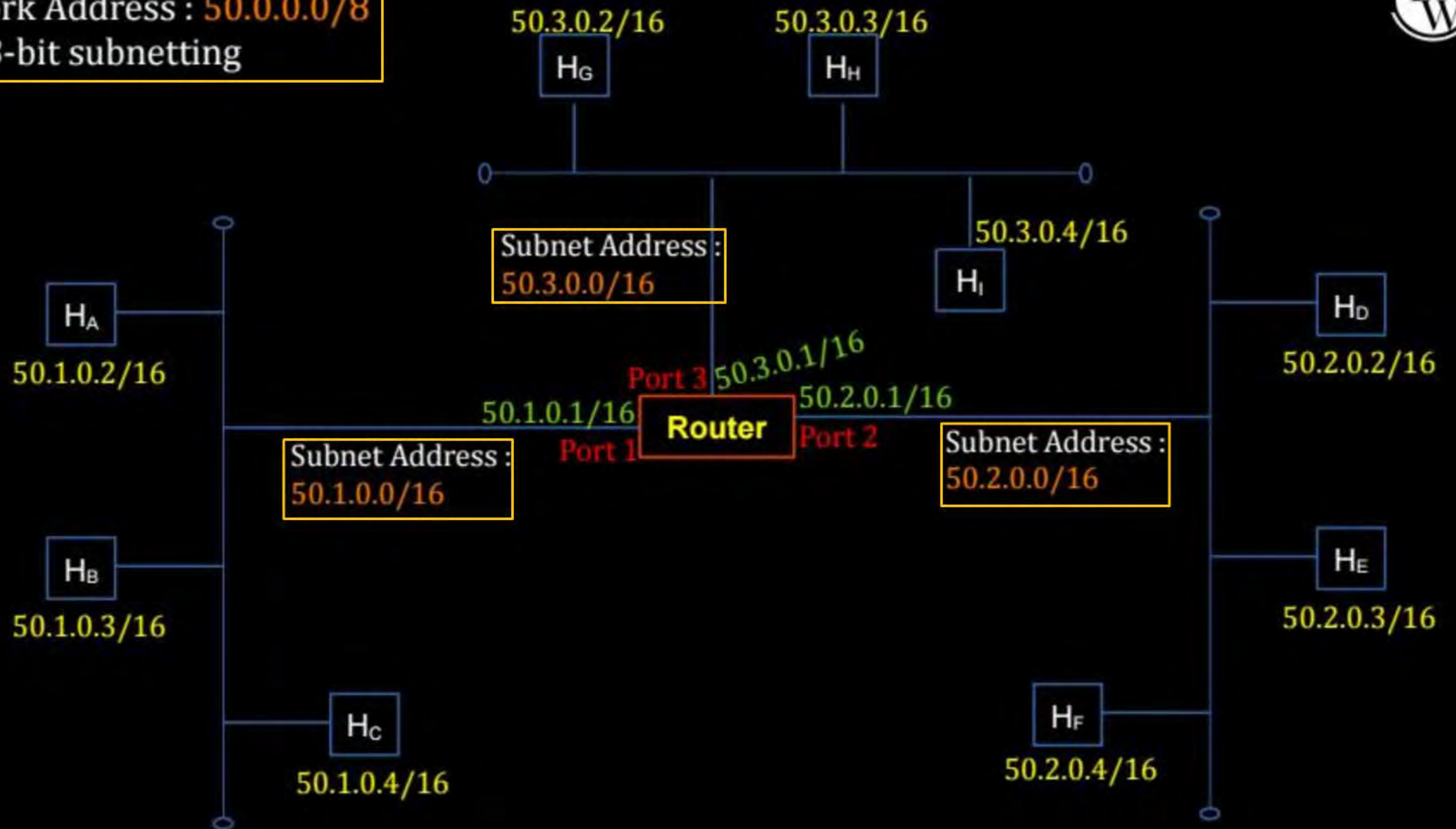
} 2^9 IP Address

CIDR Prefixes : 10.12.00000001 0.00000000 / 23

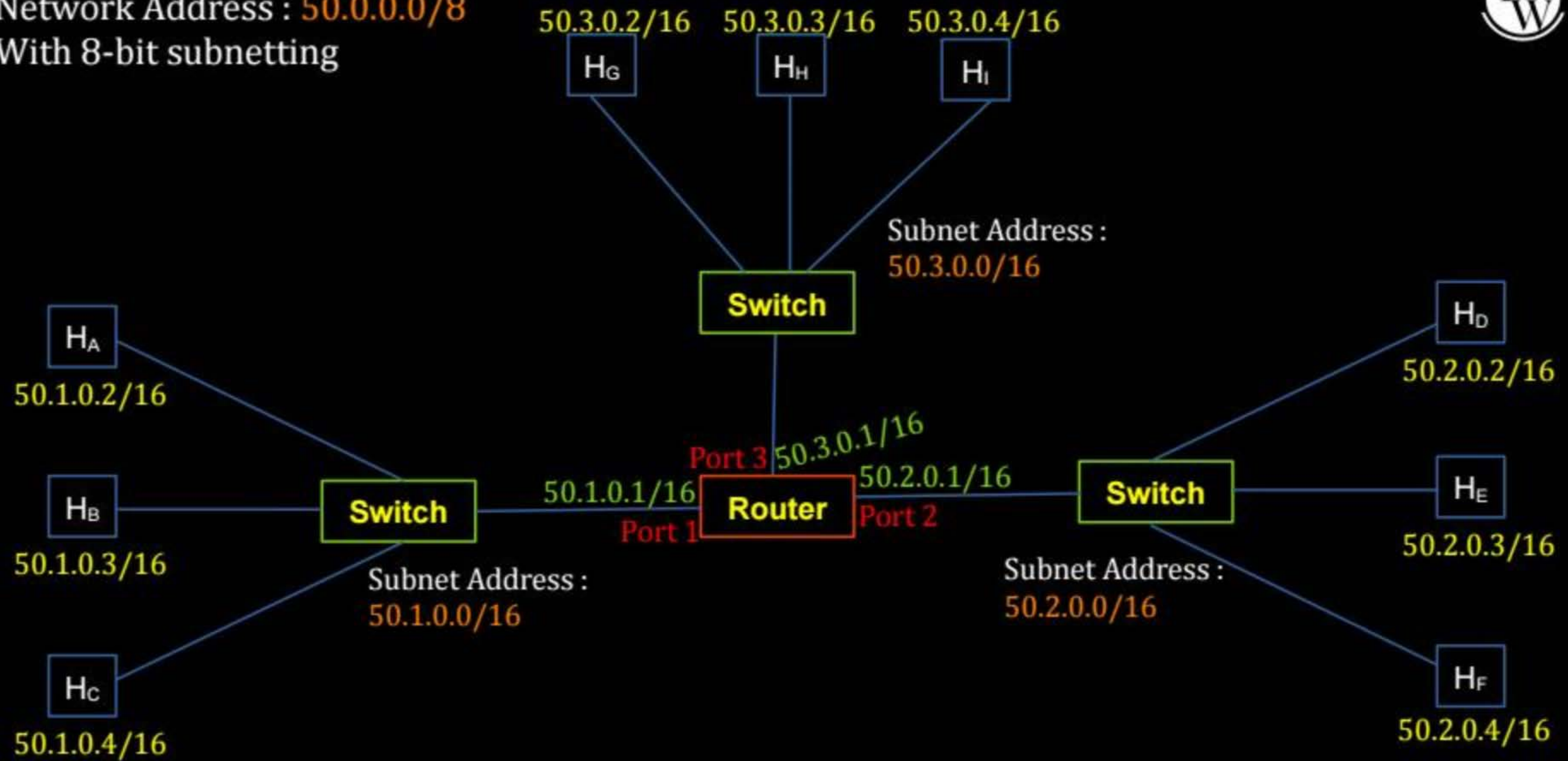
Network Address : 23 bit prefix 9 bit Host ID

10.12.2.0 / 23

Network Address : 50.0.0.0/8
With 8-bit subnetting



Network Address : 50.0.0.0/8
With 8-bit subnetting



Host A, IPv4 Configuration :-

IPv4 Address : 50 . 1 . 0 . 2

Subnet Mask : 16 or 255 . 255 . 0 . 0

Default Gateway : 50 . 1 . 0 . 1

Router forwarding table

Subnet Address	Interface ID	Next Hop
50 . 1 . 0 . 0 / 16	1	On Link
50 . 2 . 0 . 0 / 16	2	On Link
50 . 3 . 0 . 0 / 16	3	On Link

CASE II: (Part-01)

Source IP Address : 50.1.0.2

Destination IP Address : 50.2.0.2

HA

HD

50.1.0.2
255.255.0.0

50.1.0.0

50.2.0.2
255.255.0.0

50.2.0.0

- Host A (source host) finds destination host IP (Host D) belongs to different subnet
- Host A uses ARP Protocol to find MAC Address of default gateway [50.1.0.1]
- Host A send frame to Router
[The frame encapsulates the IP datagram]

Source MAC Address : Host A MAC Address

Destination MAC Address : Router MAC Address

Frame₁

CASE II : (Part-02)

Source IP Address : 50.1.0.2

Destination IP Address : 50.2.0.2

$$\begin{array}{r} 50.2.0.2 \\ 255.255.0.0 \\ \hline 50.2.0.0 \end{array}$$

- Router receives the IP datagram
- Finds destination host IP (Host D) belongs to subnet connected via port 2
- Router uses ARP Protocol to find MAC Address of destination host
- Router send frame to Host D
[The frame encapsulates the IP datagram]

Source MAC Address : Router MAC Address

Destination MAC Address : Host D MAC Address

Frame₂

CASE III :

$$\begin{array}{r} 50.1.255.255 \\ 255.255.0.0 \\ \hline 50.1.0.0 \end{array}$$

Source IP Address : 50.1.0.2 HA

Destination IP Address : 50.1.255.255

- Host A (source host) finds destination host IP belongs to same subnet
- Host A uses ARP, ARP give broadcast MAC Address
- Host A broadcast the frame into same subnet
[The frame encapsulates the IP datagram]

Source MAC Address : Host A MAC Address

Destination MAC Address : Broadcast MAC Address
[FF:FF:FF:FF:FF:FF]

CASE IV :

Source IP Address : 50.1.0.2 IHA

Destination IP Address : 255.255.255.255

- Host A (source host) finds destination host IP is limited broadcast
- Host A uses ARP, ARP give broadcast MAC Address
- Host A broadcast the frame into same subnet
[The frame encapsulates the IP datagram]

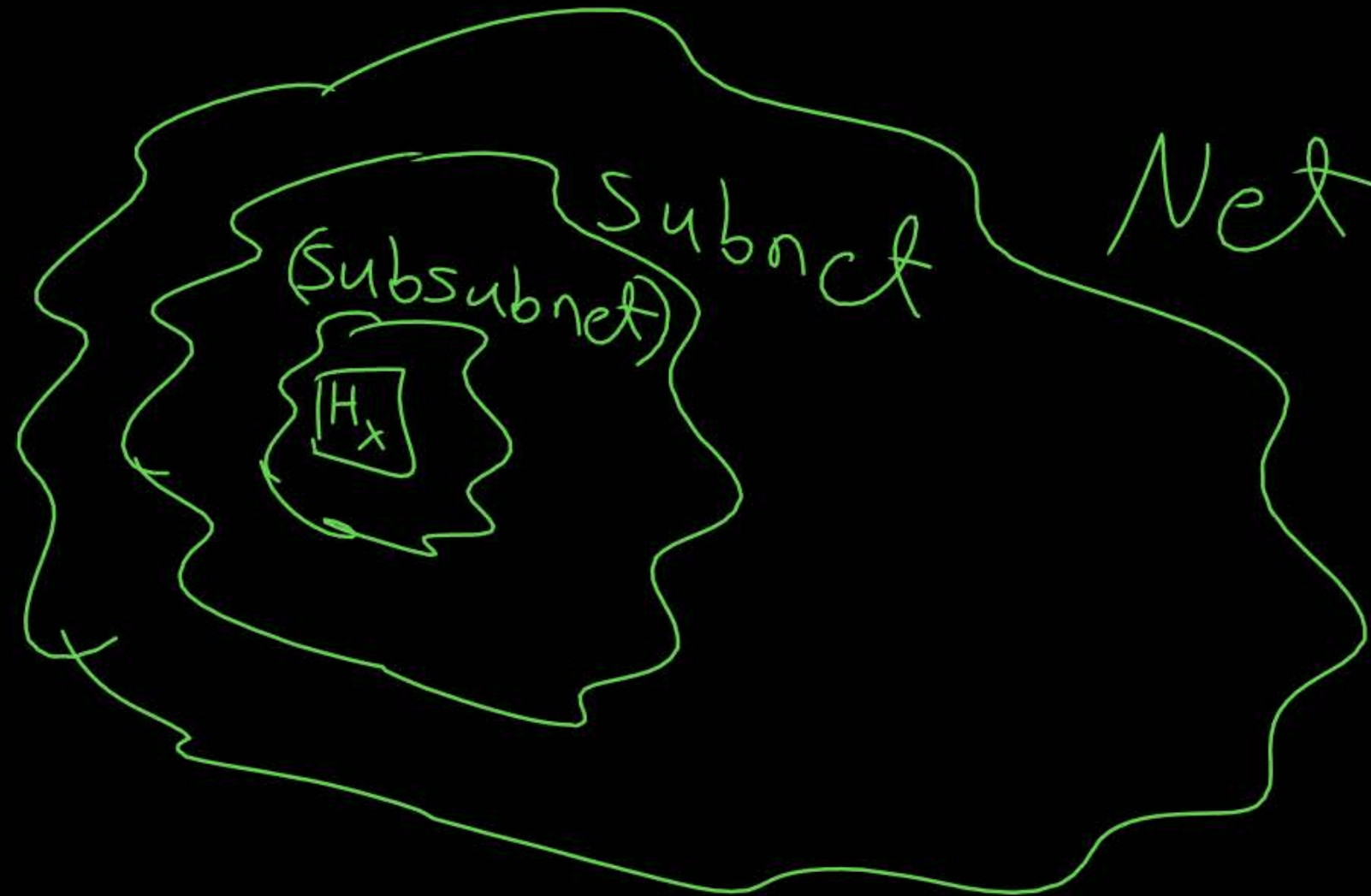
Source MAC Address : Host A MAC Address

Destination MAC Address : Broadcast MAC Address
[FF:FF:FF:FF:FF:FF]

CASE II

Source IP Add = 50.1.0.2

Dest. IP Add = 50.255.255.255,
Network Broadcast Add.



[MCQ]

[GATE-CS-2006] [2 Mark]



#Q. Two computers C1 and C2 are configured as follows. C1 has IP address "203 . 197 . 2 . 53" and netmask "255 . 255 . 128 . 0". C2 has IP address "203 . 197 . 75 . 201" and netmask "255 . 255 . 192 . 0". Which one of the following statements is true ?

- ☒ A C1 and C2 both assume they are on the same network
- ☒ B C2 assumes C1 is on same network, but C1 assumes C2 is on a different network
- ☒ C C1 assumes C2 is on same network, but C2 assumes C1 is on a different network
- ☒ D C1 and C2 both assume they are on different networks

Ans: C



Host C₁

[IP₁] : 203 . 197 . 2 . 53
[Netmask₁] : 255 . 255 . 128 . 0

Net. Add.₁ : 203 . 197 . 0 . 0

IP₂ : 203 . 197 . 75 . 201
Netmask₁ : 255 . 255 . 128 . 0

Net. Add. : 203 . 197 . 0 . 0

(Result₁)

Host C₂

[IP₂] : 203 . 197 . 75 . 201
[Netmask₂] : 255 . 255 . 192 . 0

Net. Add.₂ : 203 . 197 . 64 . 0

IP₁ : 203 . 197 . 2 . 53
Netmask₂ : 255 . 255 . 192 . 0

Net. Add. : 203 . 197 . 0 . 0

Result₂

Net Add $\rightarrow 203.197.0.0$
Mask $\rightarrow 255.255.128.0$



C₁

203.197.2.53

Subnet

Add = 203.197.64.0

Mask = 255.255.192.0

C₂

203.197.75.201

[MCQ]

H.W.

IISC

[GATE-IT-2008] [2 Mark]



#Q. Host X has IP address 192 . 168 . 1 . 97 and is connected through two routers R1 and R2 to another host Y with IP address 192 . 168 . 1 . 80, Router R1 has IP addresses 192 . 168 . 1 . 135 and 192 . 168 . 1 . 110, R2 has IP addresses 192 . 168 . 1 . 67 and 192 . 168 . 1 . 155, the netmask used in the network is 255 . 255 . 255 . 224;

Which IP address should X configure its gateway as?

A 192 . 168 . 1 . 67

B 192 . 168 . 1 . 110

C 192 . 168 . 1 . 135

D 192 . 168 . 1 . 155

[MCQ] *H. W.*

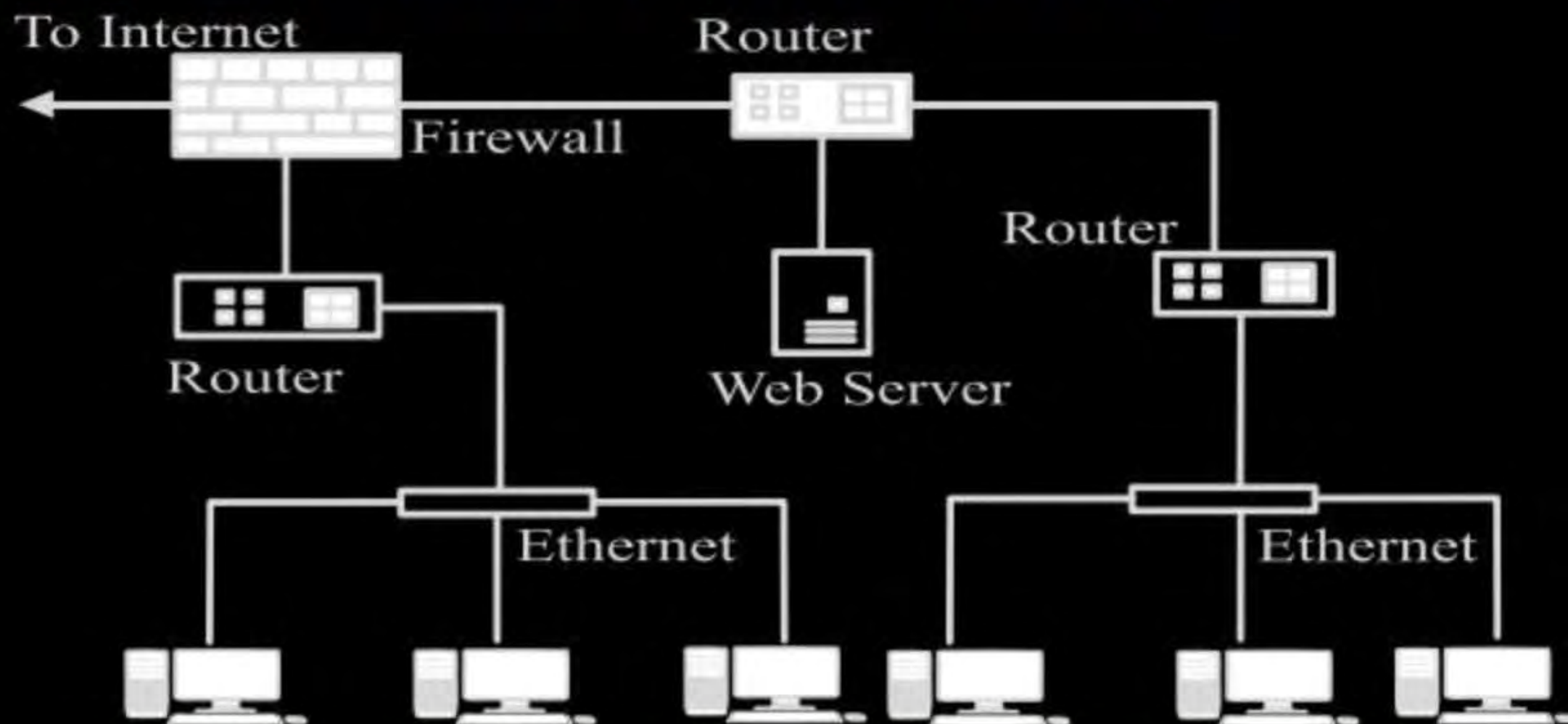
[GATE-IT-2008] [2 Mark]



#Q. Given the information in previous question, how many distinct subnets are guaranteed to already exist in the network?

- A 1
- B 2
- C 3
- D 6

#Q. Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below.



What is the number of subnets inside the enterprise network?

A

3

B

12

C

6

D

8

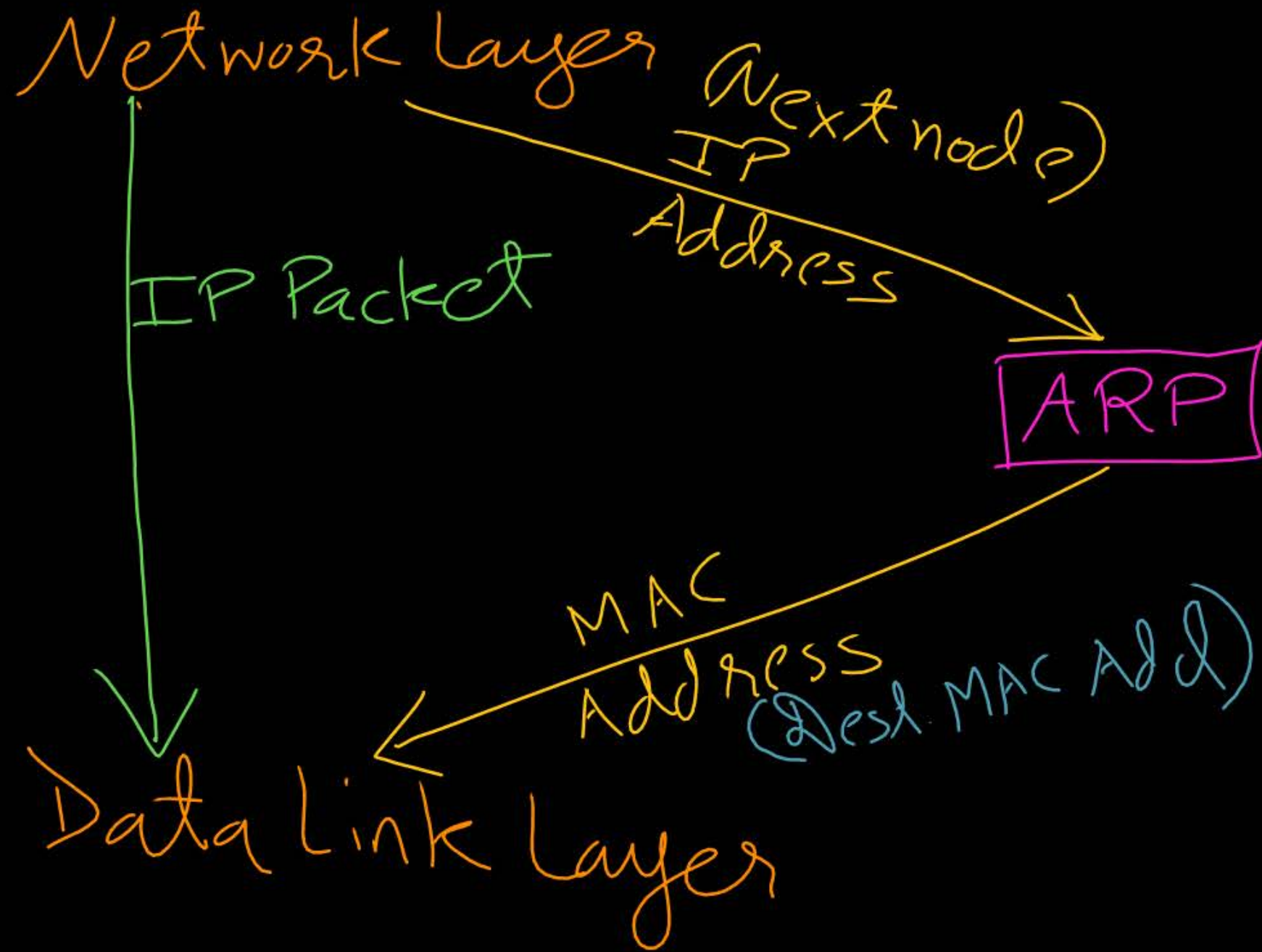


Topic : ARP



- ARP : Address Resolution Protocol
- Data Link Layer Protocol / *Network Layer Protocol*
- Used to find MAC address of device from IP address
- Map logical address into corresponding physical address
(Reverse is not true)





[MCQ]

117 → D

[GATE-CS-2005] [1 Mark]



#Q. The address resolution protocol (ARP) is used for :

- ☒ A Finding the IP address from the DNS
- ☒ B Finding the IP address of the default gateway
- ☒ C Finding the IP address that corresponds to a MAC address ⇒ RARP
- ☒ D Finding the MAC address that corresponds to an IP address

Ans: D

[MCQ]

IIT-R

[GATE-2025] [1 Mark]



#Q. Consider the following statements :

- (i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address.
- (ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S.

Which ONE of the following is CORRECT?

- A Both (i) and (ii) are TRUE
- B (i) is TRUE and (ii) is FALSE
- C (i) is FALSE and (ii) is TRUE
- D Both (i) and (ii) are FALSE

(i) is true or false?

TRUE



Topic : ARP Packet

⇒ ARP Query and Reply



- ARP Packet always encapsulated into payload field of "Frame"
- ARP Query (Request) always broadcast
[Destination MAC address = FF:FF:FF:FF:FF:FF (All 48 bits are one)
Broadcast MAC address]
- ARP Reply always unicast
[Destination MAC address = MAC address of Host who raised ARP query]

[MCQ]

IIT-B

[GATE-2021][1 Mark]



#Q. Consider the following two statements:

S_1 : Destination MAC address of an ARP reply is a broadcast address. FALSE

S_2 : Destination MAC address of an ARP request is a broadcast address. TRUE

Which one of the following choices is correct ?

A Both S_1 and S_2 are true.

B S_1 is true and S_2 is false.

✓ C S_1 is false and S_2 is true.

D Both S_1 and S_2 are false.

Ans: C



2 mins Summary



Topic

IP Configuration ✓

Topic

ARP



THANK - YOU

